

RoboCam Stentor Photobehavior Assay Protocol

Adapted from a design document our PI provided (v1.0, July 2026; kept in a local reference library, not committed here for copyright reasons). This is our own restructured summary of that protocol, cross-referenced against `WAVELENGTH_SELECTION_GUIDE.md`, which it independently corroborates: primary stimulus 610/620 nm, secondary 480/540 nm, and **680 nm explicitly as a comparison/control wavelength, not the main stimulus** — the same conclusion our own literature synthesis reached.

The three behaviors, one experiment each

Behavior	Core question	Stimulus geometry	Primary metric	Best RoboCam mode
Step-up photophobic response	Does a sudden light increase trigger stop/reversal/turn?	Whole-field temporal pulse, or dark-to-light edge	Response probability, latency, reversal duration, turn angle	High-frame-rate single-well event assay
Phototaxis	Does <i>Stentor</i> move directionally relative to a light field?	Half-field, side illumination, or smooth gradient along the light axis	Occupancy index and velocity projection	Flat chamber, stable spatial light field
Photokinesis	Does light alter swimming speed independent of direction?	Uniform full-field illumination at a defined intensity	Speed ratio, light vs. baseline	24-well screen with replicated wells

Recommended order: photophobic response first, then phototaxis, then photokinesis. Phototaxis results are ambiguous on their own — dark-side accumulation in a half-field assay can be produced by edge-triggered avoidance (photophobic response) rather than true directional steering, so it needs the photophobic assay validated first as a baseline to interpret against.

Design principles

- **Separate stimulus geometry from behavior class.** Temporal steps for photophobia, gradients/half-fields for taxis, uniform light for kinesis. Conflating these risks mislabeling a photophobic edge artifact as “phototaxis.”
- **Synchronize video and LED timing.** Record LED-on timestamps in the same data stream as the video frame index (or add a visible timing-indicator LED). Without this, latency and event scoring can’t be trusted.
- **Use shallow, optically flat chambers.** Flatten the meniscus, constrain depth, keep *Stentor* in a single focal layer — losing focus through depth is usually a bigger problem than insufficient magnification, since *Stentor* is already large (approaching/exceeding 1 mm extended).
- **Calibrate irradiance at the sample plane.** Report wavelength, irradiance, pulse duration, duty cycle, and chamber geometry — PWM percentage or supply voltage alone isn’t comparable across builds.
- **Use non-actinic imaging light.** Dim IR (850/940 nm) or very dim red/far-red for tracking, verified with a tracking-light-only control to confirm it doesn’t itself elicit a response.
- **Block and randomize.** Randomize well order, wavelength order, and intensity sequence within culture-day blocks, so time/evaporation/fatigue/culture condition doesn’t masquerade as a light effect.

Stimulus wavelengths

Channel	Purpose	Notes
610 or 620 nm	Primary stimulus	Most important channel for photophobic and phototaxis assays
540 or 480 nm	Secondary action-spectrum point	Useful after the main assay works; don't start here
660/680 nm	Long-red comparison	Expected to be less effective than 610/620 nm; used as a control
850/940 nm	Imaging only	Verify with a tracking-light-only control

Use a constant-current LED driver rather than driving high-power LEDs directly from a microcontroller; have RoboCam control the driver enable pin or a logic-level MOSFET gate, and log the actual command waveform (ideally alongside a photodiode signal from the sample plane during validation).

Chamber design

- **First assay:** a shallow chamber, not an open 24-well meniscus — glass-bottom or optically clear well with a removable spacer/gasket and a top cover film/cover slip. Target depth 0.5-1.5 mm (deep enough to avoid compressing cells, shallow enough for focus and intensity uniformity); planar area 8-15 mm for single-well tracking.
- **24-well plates** are appropriate for photokinesis screens and replicated intensity/wavelength tests, but raw deep-well distributions shouldn't be interpreted as taxis unless the light field and depth are characterized — curved meniscus and wall effects can dominate the distribution.
- For one-well-at-a-time scanning, randomize well order so plate position and elapsed time aren't confounded with condition.

Assay 1 — Step-up photophobic response (start here)

Recommended as the first RoboCam validation assay: gives a time-locked event with clear behavioral labels.

- **Pulse durations:** 0.2, 0.5, 1.0, 2.0 s. **Intensity:** log-spaced pilot series at the sample plane, to fit dose-response curves and estimate an EC50-like threshold. **Inter-trial interval:** at least 5-10 s (longer if repeated stimulation shows adaptation). **Frame rate:** 60 fps minimum, 120 fps preferred for latency/reversal dynamics.
- **Event definitions:** baseline speed = median speed -2.0 to -0.2 s before LED onset; response window = 0.05-1.5 s after onset; stop event = speed drops below 20-30% of baseline for ≥ 2 -3 consecutive frames; reversal event = negative displacement relative to pre-stimulus heading for ≥ 100 -150 ms; latency = time from LED onset to first stop/reversal frame; turn angle = angle between mean heading before stimulus and mean heading after recovery.
- **Output metrics:** response probability, median latency/IQR by wavelength and irradiance, reversal duration and backward distance, turn-angle distribution, dose-response curve (probability vs. log irradiance), adaptation curve across repeated pulses.

Assay 2 — Phototaxis / directional bias (only after Assay 1 works)

- **Half-field chamber:** mask/project half the chamber dark, half bright at 610/620 nm. Readout: dark-side occupancy over time and edge-crossing response — but this conflates true steering with photophobic boundary avoidance.
- **Smooth gradient:** diffuser + neutral-density gradient, wedge mask, or side illumination across the chamber. Readout: velocity projection along the gradient axis — better isolates steering/taxis, but needs a calibrated light field.

- **Metrics:** occupancy index = $(N_{\text{dark}} - N_{\text{light}})/N_{\text{total}}$, reported over time, not just at the end; directional bias = mean velocity projected away from the light source, divided by speed (near +1 = moving away, near 0 = no bias); edge-crossing response probability; dwell time per zone, normalized by zone area.
- **Caution:** don't call a result "true phototaxis" from dark-side accumulation alone — pair it with velocity-vector analysis and the photokinesis control. Avoid bubbles, curved meniscus, and wall shadows, which generate artificial gradients and trapping zones.

Assay 3 — Photokinesis / speed modulation

The best fit for 24-well RoboCam scans, since uniform full-field illumination is easier to reproduce across wells than a gradient.

- **Windows:** baseline 30-60 s before light; acute onset 0-2 s after light-on (inspect, but don't use as the main metric — photophobic events can contaminate speed here); sustained light 10-60 s after light-on (primary photokinesis estimate); recovery 30-60 s after light-off (reversibility/fatigue).
- **Photokinesis index** = median speed during sustained light / median baseline speed. Use a paired design — each well/cell as its own baseline control. Exclude tracks that hit walls, attach to substrate, or leave focus for more than a defined fraction of the window.
- **First screen:** 610/620 nm at low/medium/high irradiance, plus a no-light control and a 680 nm comparison.

Controls and confound checks

Control	Purpose	Pass criterion
No-actinic sham	Camera/vibration/software-trigger artifacts	No increase in stop/reversal events at sham time
Tracking light only	Confirms IR/far-red imaging is behaviorally silent	Baseline speed/event frequency match dark control
680 nm comparison	Separates wavelength-specific sensitivity from generic brightness/heating	Lower or shifted response vs. 610/620 nm at matched irradiance
Heat control	Tests whether LED warming changes swimming speed	Temperature rise below a predefined limit; speed not explained by heat alone
Spatial mask blank	Tests whether masks create flow/shadow/focusing artifacts	No preferential accumulation without actinic contrast
Fixed particle video	Detects stage drift and optical distortion	No apparent movement in fixed particles during LED switching

Recommended workflow

Phase 0 — build and calibration: measure field uniformity (photodiode/camera flat-field per mask and LED channel); verify LED timing (film an indicator LED or log a photodiode signal alongside video); measure sample-plane temperature during the longest planned exposure; run tracking-light-only controls; record a calibration video with beads/fixed particles for pixel-to-micron scale.

Phase 1 — minimum viable experiment: load 5-20 healthy swimming cells into a shallow chamber (avoid dense cultures, disturbed cells, debris); 5-10 min acclimation/dark adaptation; 10 s baseline under non-actinic light; a 610/620 nm pulse series (0.5 s and 1.0 s at low/medium/high irradiance, 5-10 s between trials); manually score the first 20-50 events before comparing to automated scoring; repeat with sham pulses and a 680 nm comparison at matched irradiance.

Phase 2 — publication-grade assay: ≥ 3 independent culture days/batches as biological replicates; multiple wells/chambers per condition per day as technical replicates; report analyzable cells/tracks, but treat

culture day/well as the primary replicate level; randomize intensity and well order (avoid monotonic low-to-high sequences unless specifically studying adaptation); blind manual event scoring to wavelength/intensity where practical; predefine QC exclusions (stuck cells, collisions, track loss, fission-stage cells, debris).

Analysis pipeline

Preprocessing (flat-field correction, background subtraction, timestamp alignment) -> segmentation (contour/threshold tuned for large *Stentor* bodies, ignoring sub-threshold particles) -> tracking (nearest-neighbor/Kalman linking with gap-closing) -> event detection (speed/heading/displacement relative to the pre-stimulus vector) -> metrics (response probability, latency, reversal duration, speed ratio, occupancy index, directional bias) -> archiving (raw video, metadata JSON/CSV, processed tracks, analysis code version — a reproducible experiment folder per run).

Minimum metadata to record per run

Experiment (id, date, operator, RoboCam build/version, analysis version); biology (species, culture source/age, feeding status, medium, temperature); chamber (plate type, depth, gasket material, volume, cover method, well ID); optics (camera model, lens, magnification, frame rate, resolution, pixel size, imaging light wavelength/intensity); stimulus (LED wavelength, bandwidth if known, irradiance at sample plane, pulse duration, duty cycle, mask/gradient geometry); trial (baseline duration, stimulus onset frame, recovery duration, inter-trial interval, randomized order index); QC (cells loaded, cells analyzable, exclusions, notes on bubbles/debris/focus).

Statistics

- Photophobic response probability: logistic regression / generalized mixed model, wavelength/intensity as fixed effects, culture day/well as random/blocking factors.
- Latency and reversal duration: medians or robust models (distributions are often skewed and censored by frame rate).
- Photokinesis: paired speed ratios by cell or well, aggregated by well/culture day — don't treat every frame as an independent replicate.
- Phototaxis: report both occupancy and directional movement — occupancy alone is insufficient, since cells can accumulate in dark zones through photophobic boundary rejection rather than true steering.
- Report failed tracking and exclusions transparently; these assays are highly sensitive to chamber quality and culture condition.

Suggested 24-well plate layout for the initial screen

Wells	Condition	Purpose
A1-A3	No-actinic sham, imaging only	Baseline / tracking-light control
A4-A6	610/620 nm low irradiance	Lower end of dose-response
B1-B3	610/620 nm medium irradiance	Expected working condition
B4-B6	610/620 nm high irradiance	Saturation/adaptation check
C1-C3	680 nm matched irradiance	Long-red wavelength comparison
C4-C6	Uniform 610/620 nm sustained light	Photokinesis window
D1-D3	Half-field 610/620 nm	Phototaxis/edge-avoidance pilot
D4-D6	Recovery/dark after exposure	Carryover and fatigue check

Don't run the plate in A1-to-D6 order for one-well-at-a-time scanning — generate a randomized acquisition order instead.